Thesis ROS2 Simulation Results

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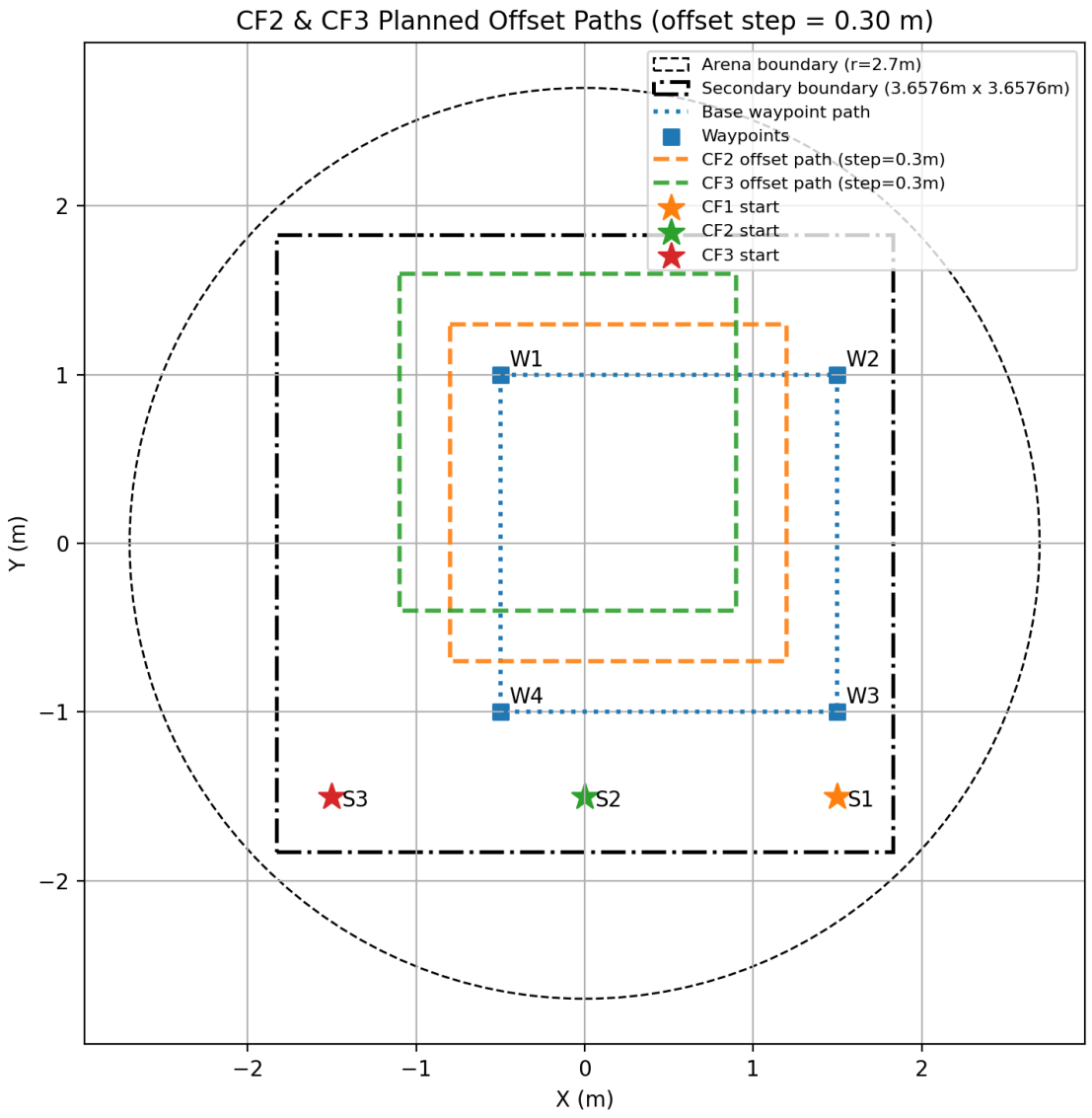
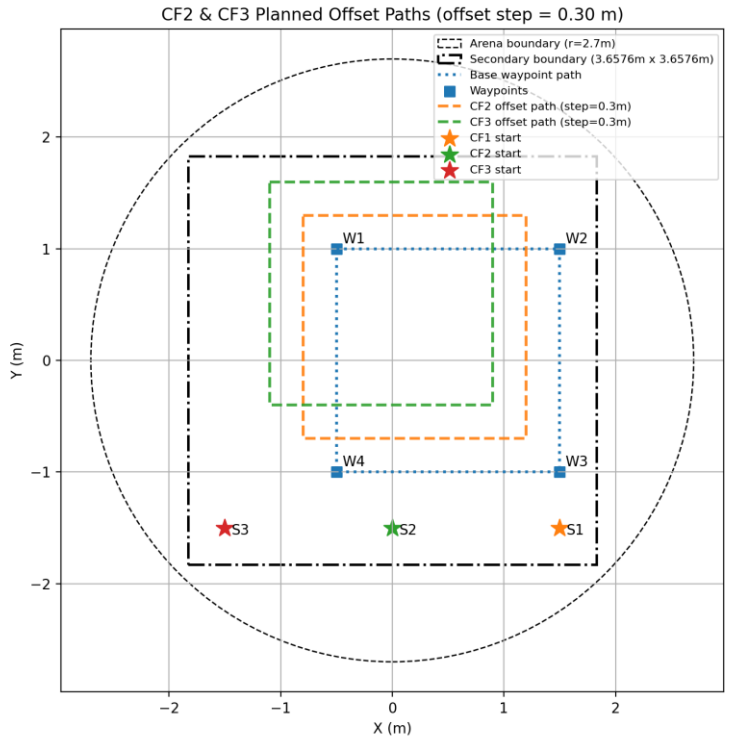
Setup

Start	Waypoints
CF1 (1.5,-1.5)	W1 (-0.5, 0.5)
CF2 (0 ,-1.5)	W2 (0.5, 0.5)
CF3 (-1.5,-1.5)	W3 (0.5,-0.5)
	W4 (- 0.5,- 0.5)

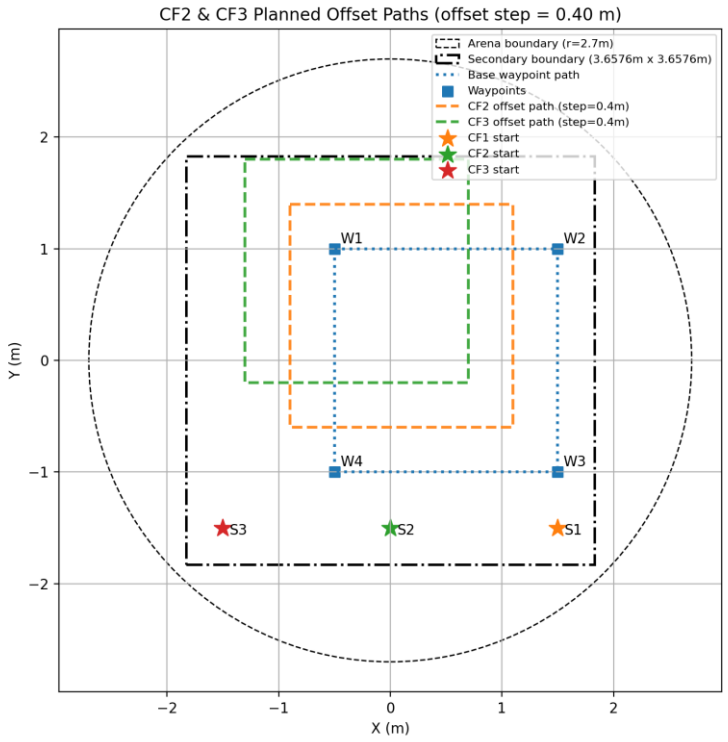
waypoint_offset = 1 x dmin



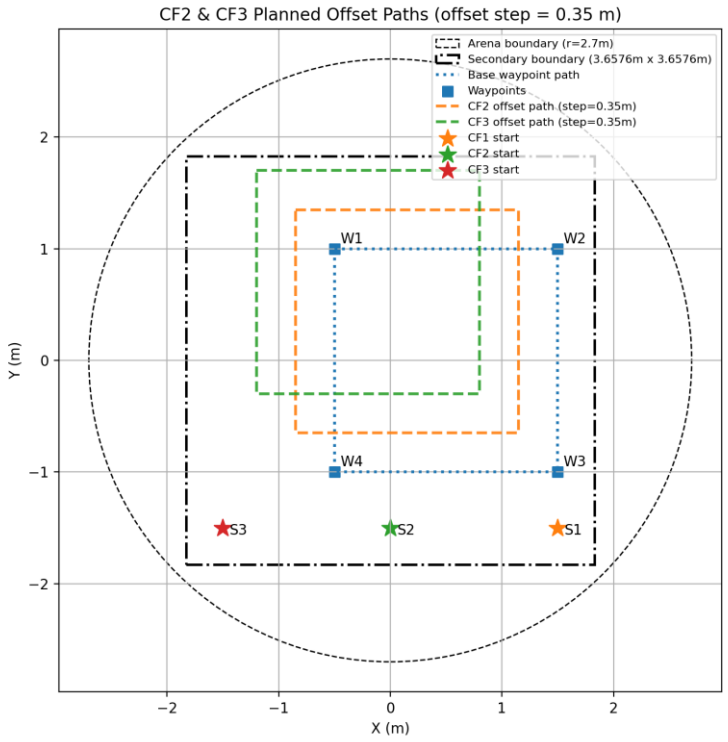
Offset paths



0.30m



0.35m



0.40m

Variables

- **Fixed**
- - Vmax = 0.20 m/s
- - z = 0.5 m
- - closeEnough = 0.30 m (tolerance radius to waypoint)
- - EPS (too close) = 0.05 m (tolerance safety/repel)

- **Variable**
- - Dmin 0.30 - 0.35 - 0.40
- - Dmax 1.00 - 1.10 - 1.20
- - Threshold 0.80 - 1.70

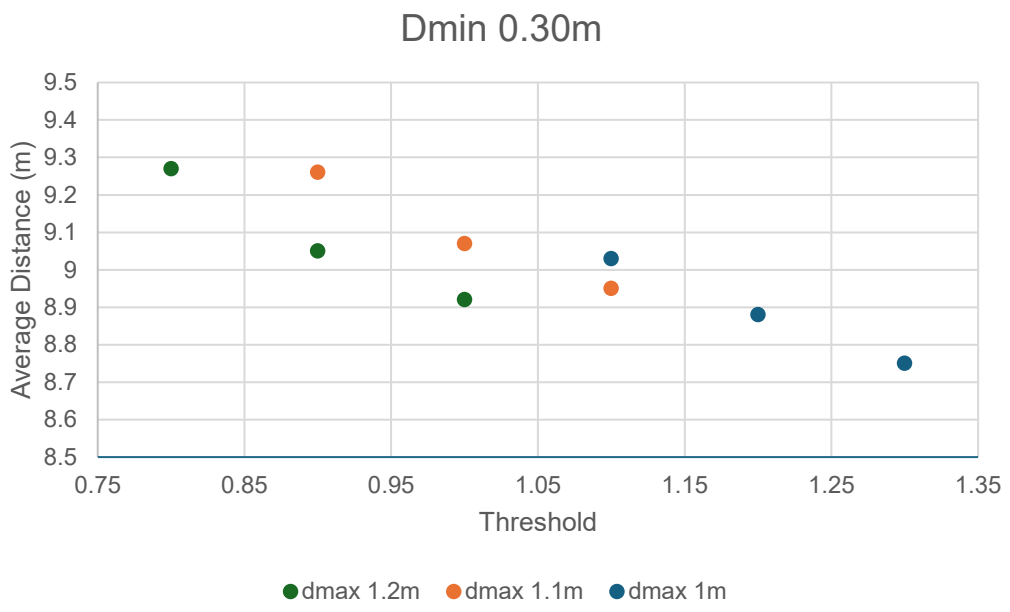
Results

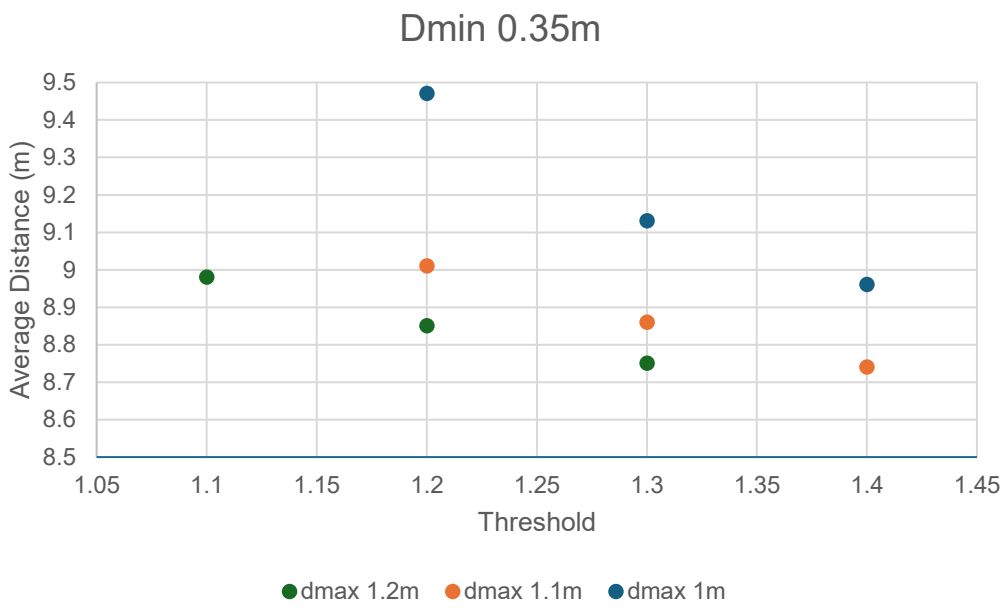
Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.3	1	1				X
		1.1	8.65	8.74	9.71	9.03
		1.2	8.4	8.61	9.62	8.88
		1.3	8.27	8.54	9.45	8.75
	1.1	0.8				x
		0.9	9.07	8.84	9.88	9.26
		1	8.75	8.75	9.71	9.07
		1.1	8.52	8.71	9.64	8.95
	1.2	0.7				X
		0.8	8.88	8.93	10	9.27
		0.9	8.71	8.76	9.69	9.05
		1	8.58	8.62	9.55	8.92

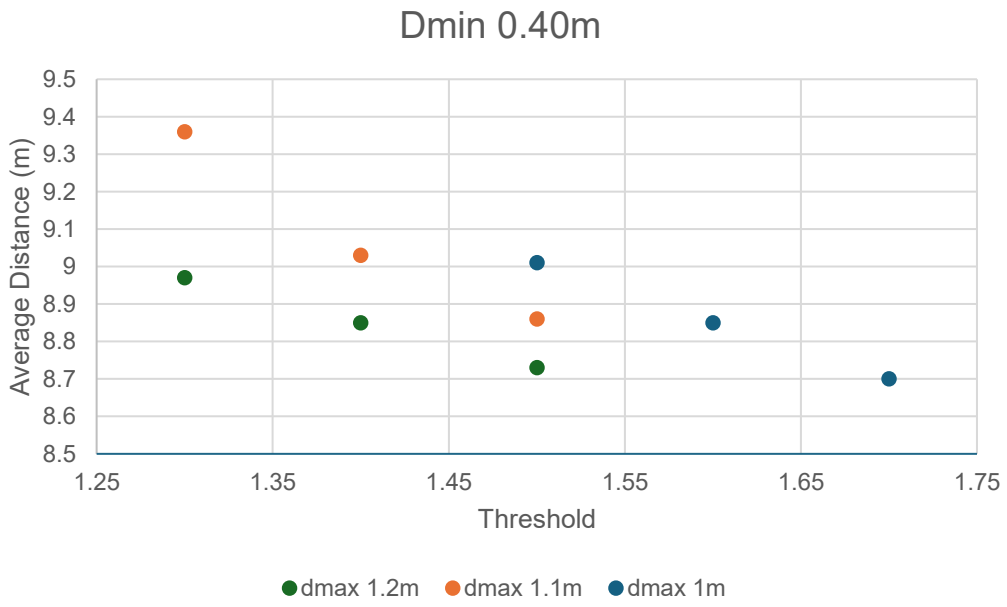
Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.35	1	1.1				x
		1.2	9.17	8.84	10.38	9.47
		1.3	8.57	8.7	10.14	9.13
		1.4	8.48	8.78	9.63	8.96
	1.1	1.1				X
		1.2	8.58	8.71	9.74	9.01
		1.3	8.37	8.71	9.5	8.86
		1.4	8.21	8.57	9.45	8.74
	1.2	1				X
		1.1	8.53	8.75	9.67	8.98
		1.2	8.35	8.63	9.58	8.85
		1.3	8.32	8.78	9.47	8.85

Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.4	1	1.4				x
		1.5	8.52	8.79	9.7	9.01
		1.6	8.27	8.7	9.59	8.85
		1.7	8.19	8.52	9.4	8.7
	1.1	1.2				X
		1.3	9	8.84	10.26	9.36
		1.4	8.39	8.76	9.93	9.03
		1.5	8.21	8.83	9.55	8.86
	1.2	1.2				X
		1.3	8.5	8.81	9.61	8.97
		1.4	8.23	8.81	9.51	8.85
		1.5	8.13	8.62	9.43	8.73

Results


Dmin 0.30m
Average Distance (m)
9.5
9.4
9.3
9.2
9.1
9
8.9
8.8
8.7
8.6
8.5
0.75
0.85
0.95
1.05
1.15
1.25
1.35
Threshold
dmax 1.2m
dmax 1.1m
dmax 1m

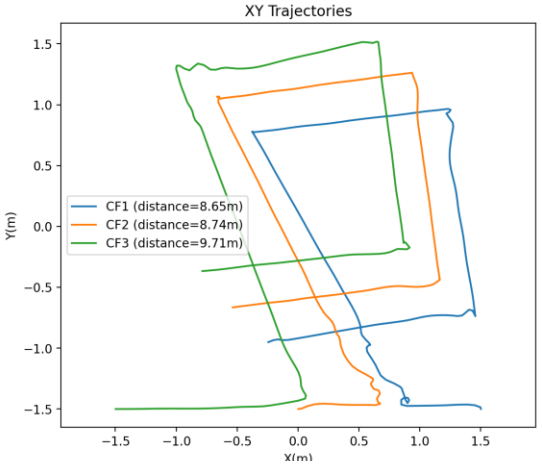

Dmin 0.35m
Average Distance (m)
9.5
9.4
9.3
9.2
9.1
9
8.9
8.8
8.7
8.6
8.5
1.05
1.1
1.15
1.2
1.25
1.3
1.35
1.4
1.45
Threshold
dmax 1.2m
dmax 1.1m
dmax 1m

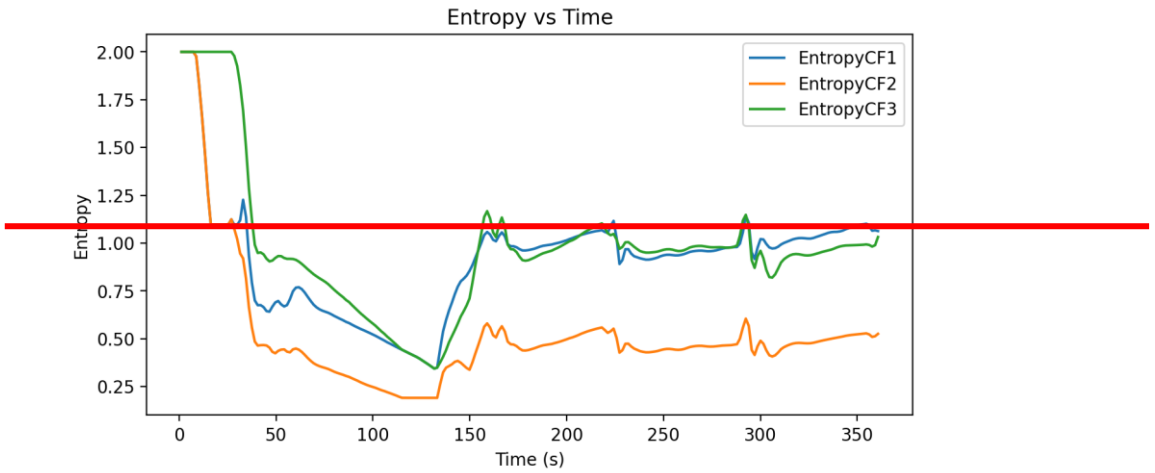

Dmin 0.40m
Average Distance (m)
9.5
9.4
9.3
9.2
9.1
9
8.9
8.8
8.7
8.6
8.5
1.25
1.35
1.45
1.55
1.65
1.75
Threshold
dmax 1.2m
dmax 1.1m
dmax 1m

Dmin 0.30

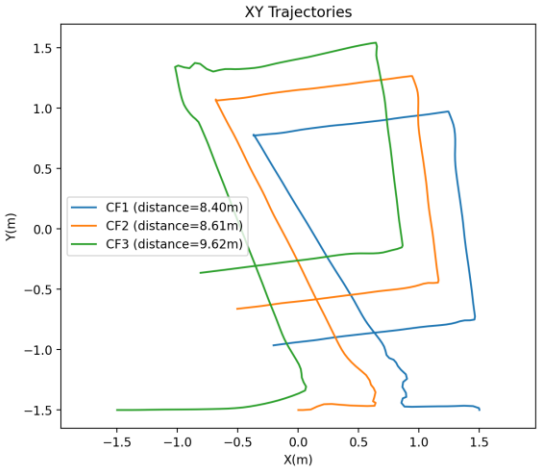
Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.3	1	1				X
		1.1	8.65	8.74	9.71	9.03
		1.2	8.4	8.61	9.62	8.88
		1.3	8.27	8.54	9.45	8.75
	1.1	0.8				x
		0.9	9.07	8.84	9.88	9.26
		1	8.75	8.75	9.71	9.07
		1.1	8.52	8.71	9.64	8.95
	1.2	0.7				X
		0.8	8.88	8.93	10	9.27
		0.9	8.71	8.76	9.69	9.05
		1	8.58	8.62	9.55	8.92

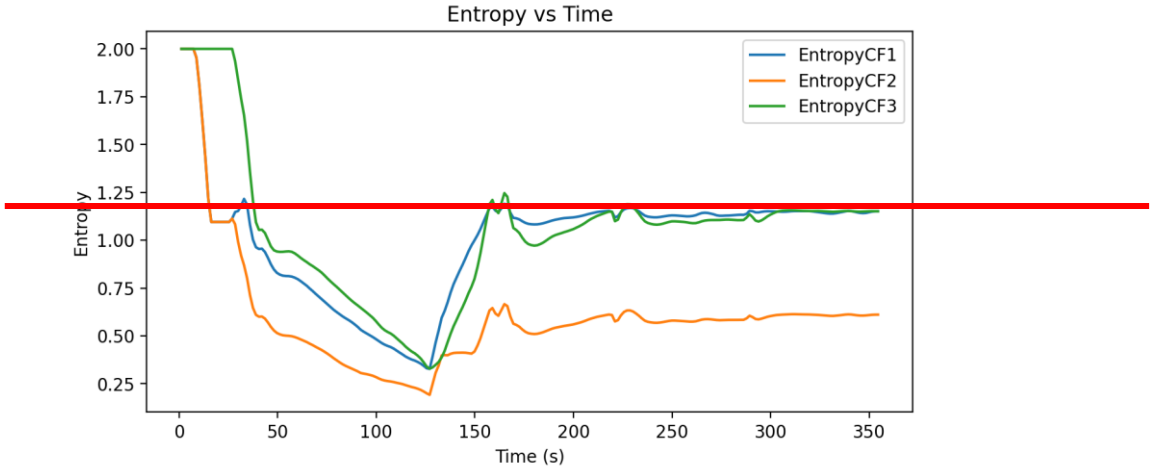
Dmin: 30cm
Dmax: **1.00m**
Threshold: 1.10



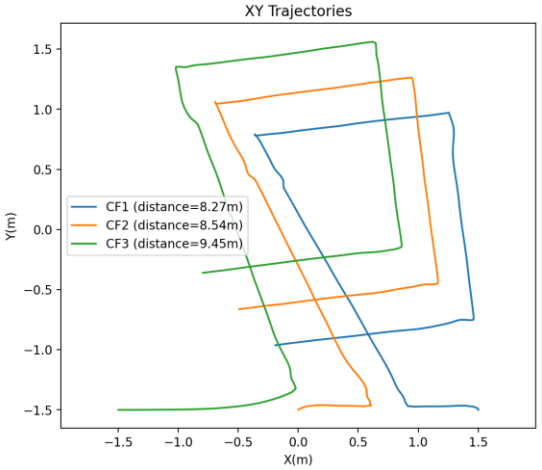


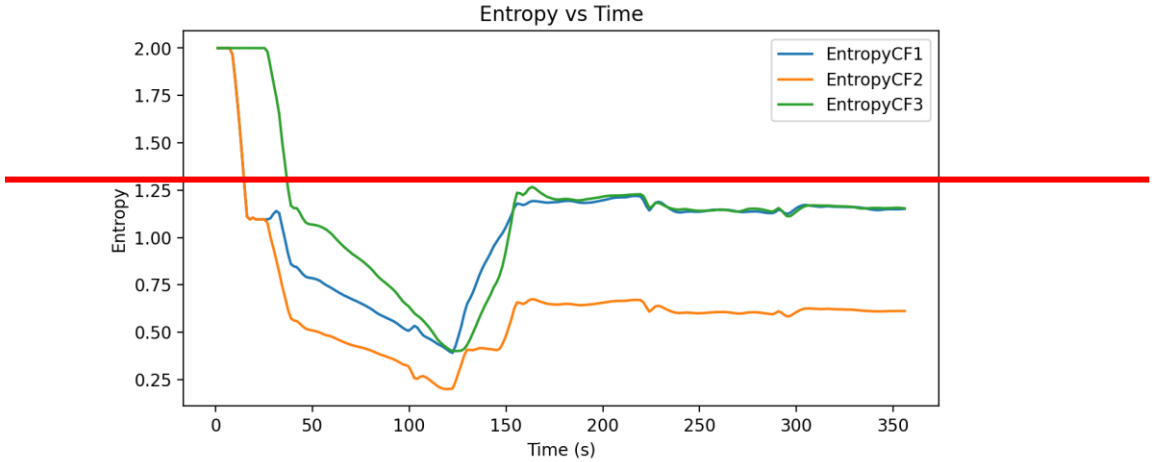
Dmin: 30cm
Dmax: **1.00m**
Threshold: 1.20



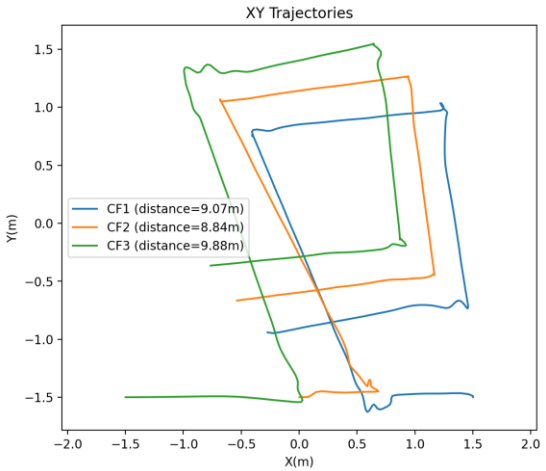


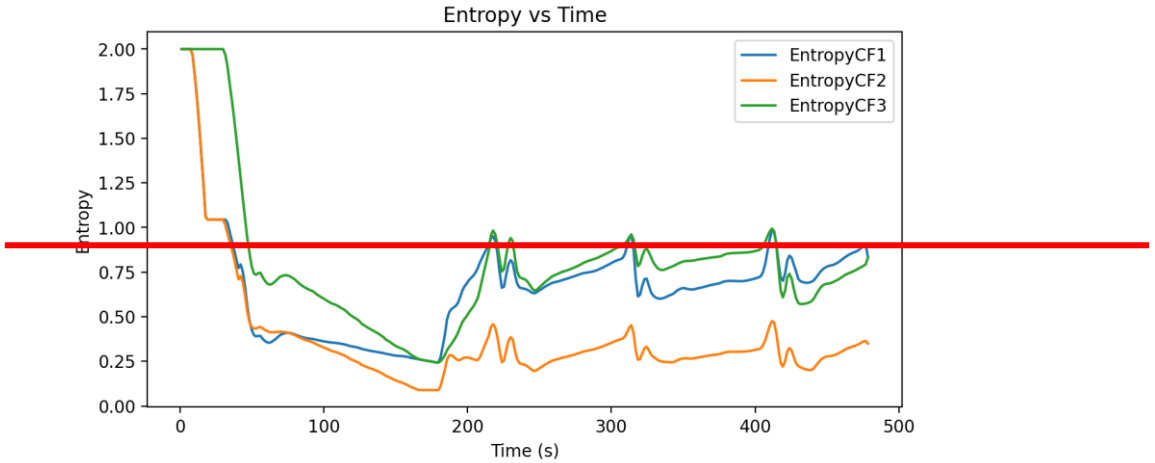
Dmin: 30cm
Dmax: **1.00m**
Threshold: 1.30



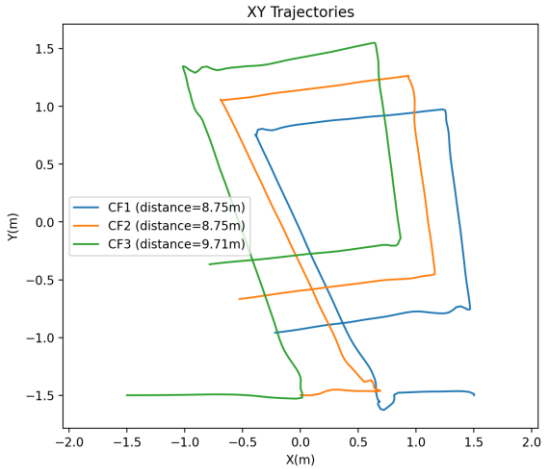


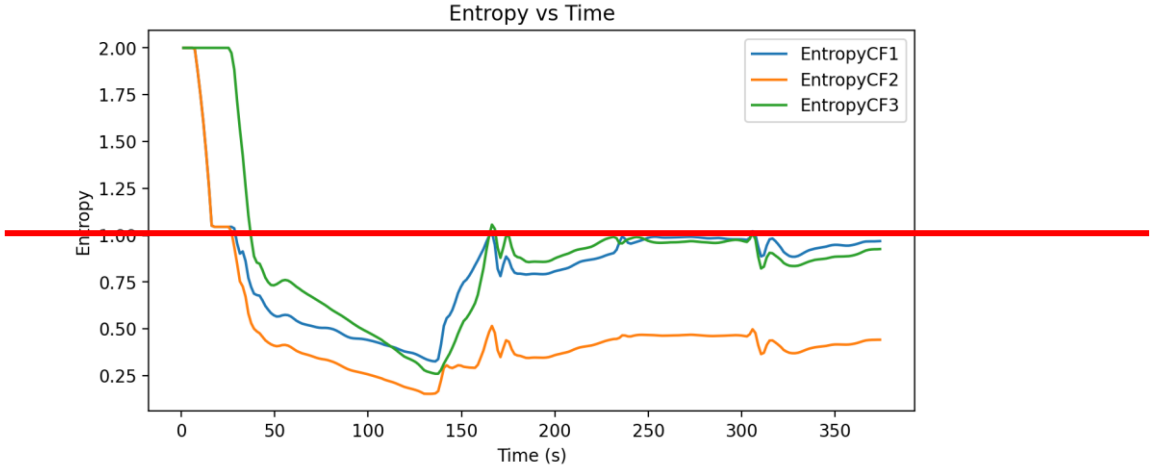
Dmin: 30cm
Dmax: **1.10m**
Threshold: 0.90



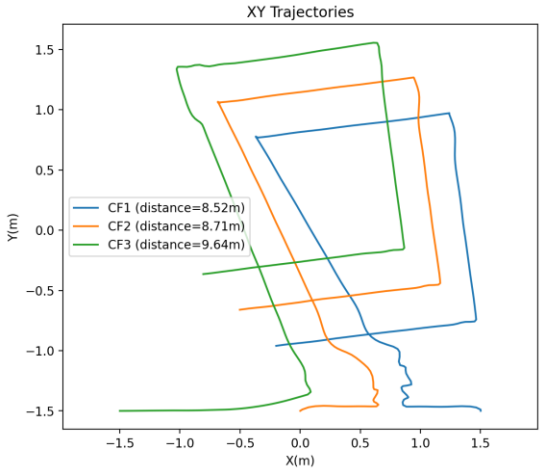


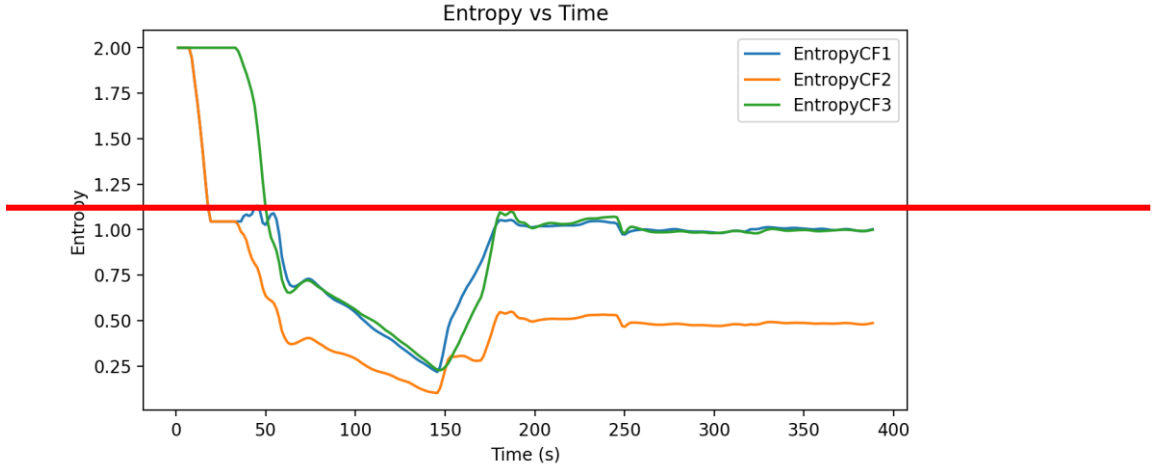
Dmin: 30cm
Dmax: **1.10m**
Threshold: 1.00



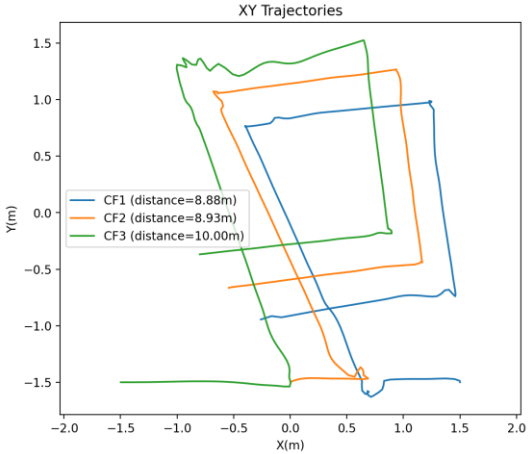


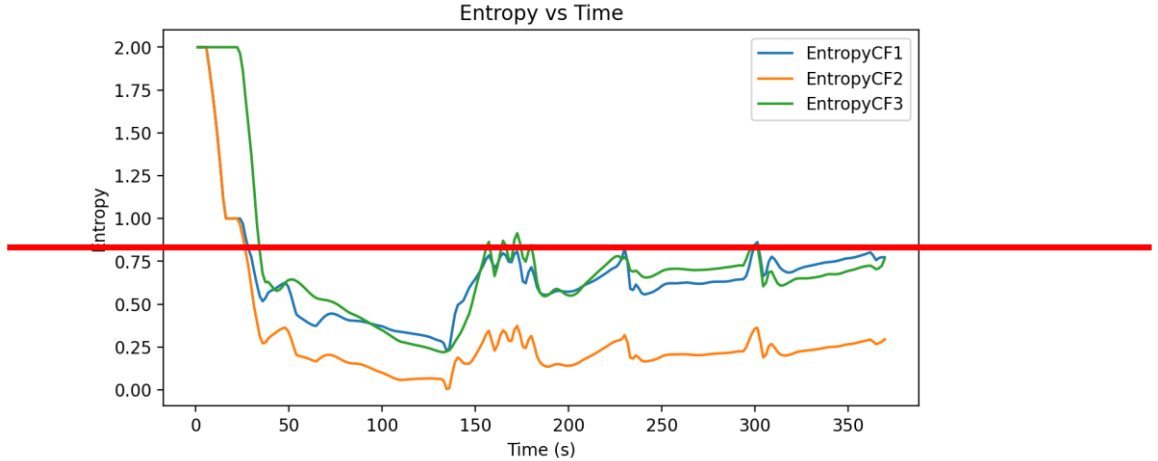
Dmin: 30cm
Dmax: **1.10m**
Threshold: 1.10



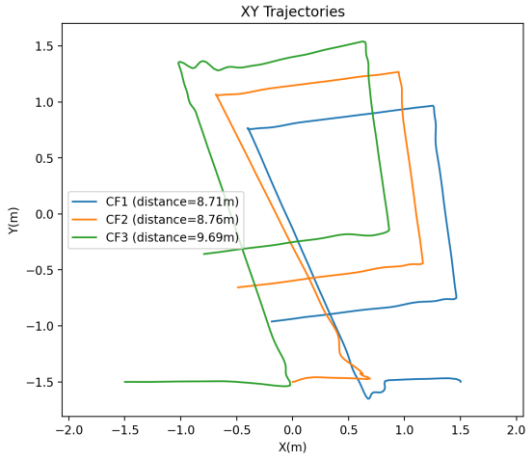


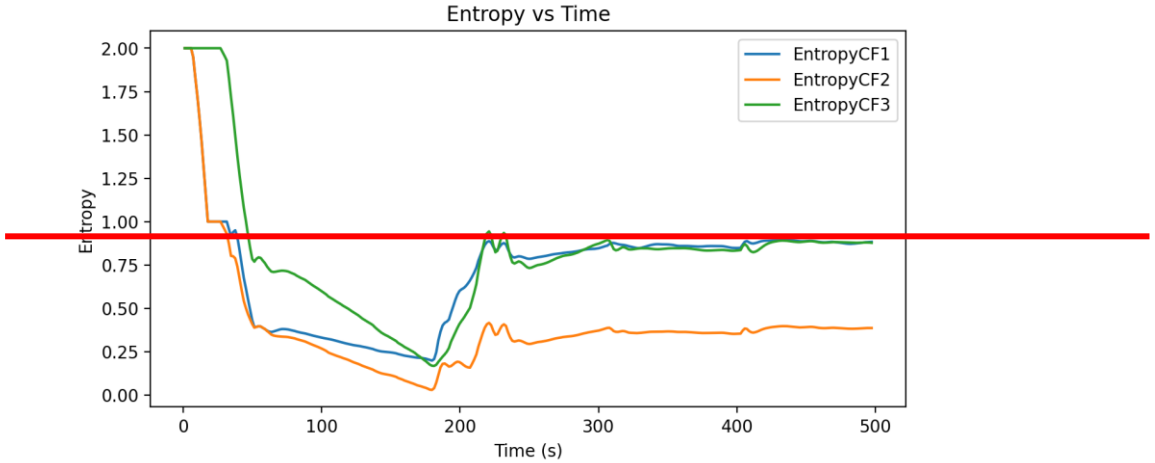
Dmin: 30cm
Dmax: **1.20m**
Threshold: 0.80



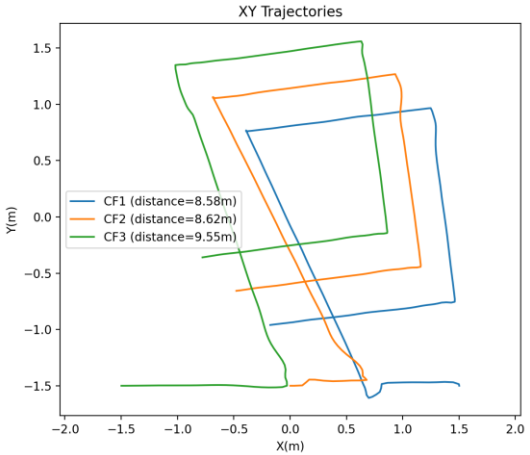


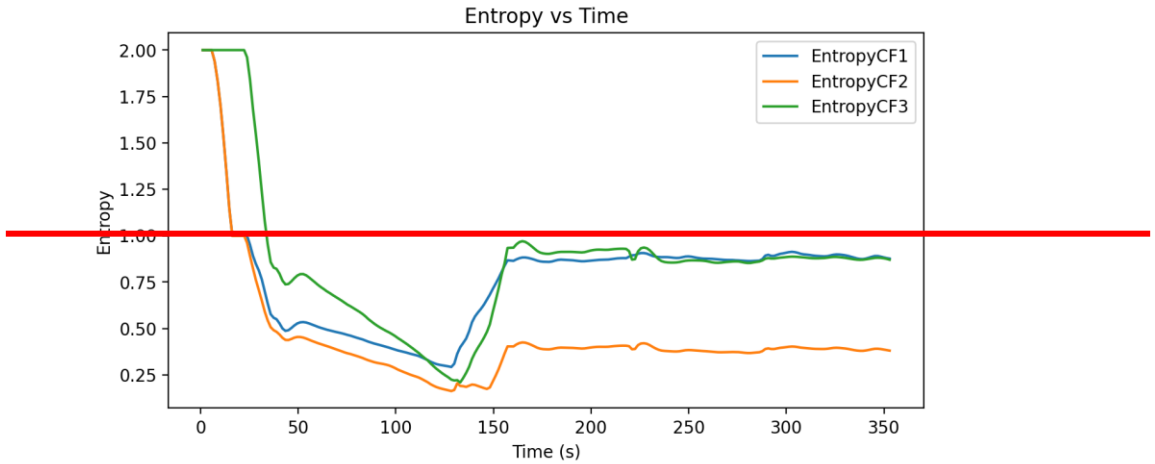
Dmin: 30cm
Dmax: **1.20m**
Threshold: 0.90





Dmin: 30cm
Dmax: **1.20m**
Threshold: 1.00

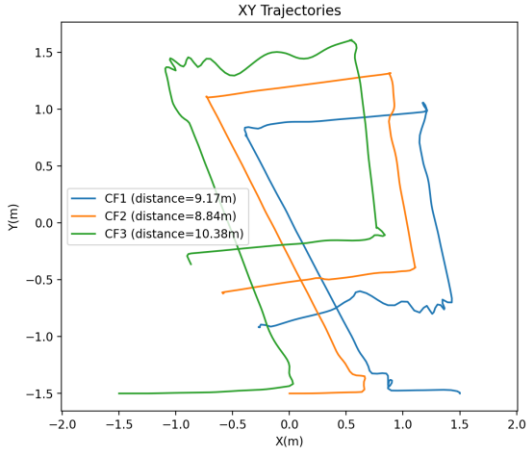


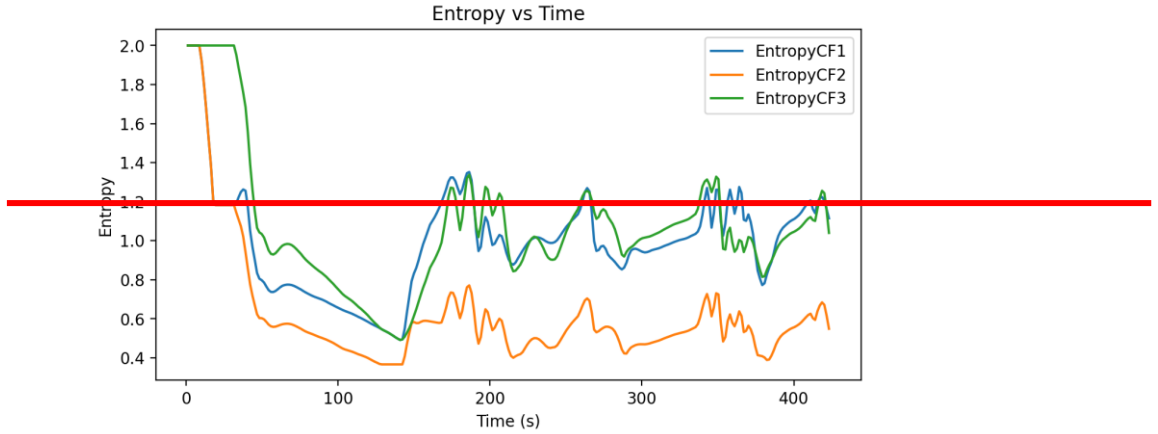


Dmin 0.35

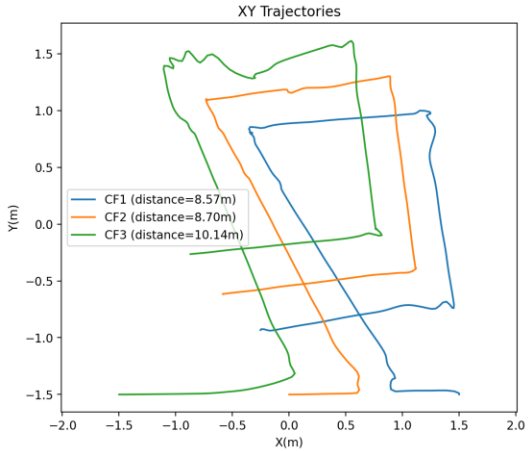
Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.35	1	1.1				x
		1.2	9.17	8.84	10.38	9.47
		1.3	8.57	8.7	10.14	9.13
		1.4	8.48	8.78	9.63	8.96
	1.1	1.1				X
		1.2	8.58	8.71	9.74	9.01
		1.3	8.37	8.71	9.5	8.86
		1.4	8.21	8.57	9.45	8.74
	1.2	1				X
		1.1	8.53	8.75	9.67	8.98
		1.2	8.35	8.63	9.58	8.85
		1.3	8.32	8.78	9.47	8.85

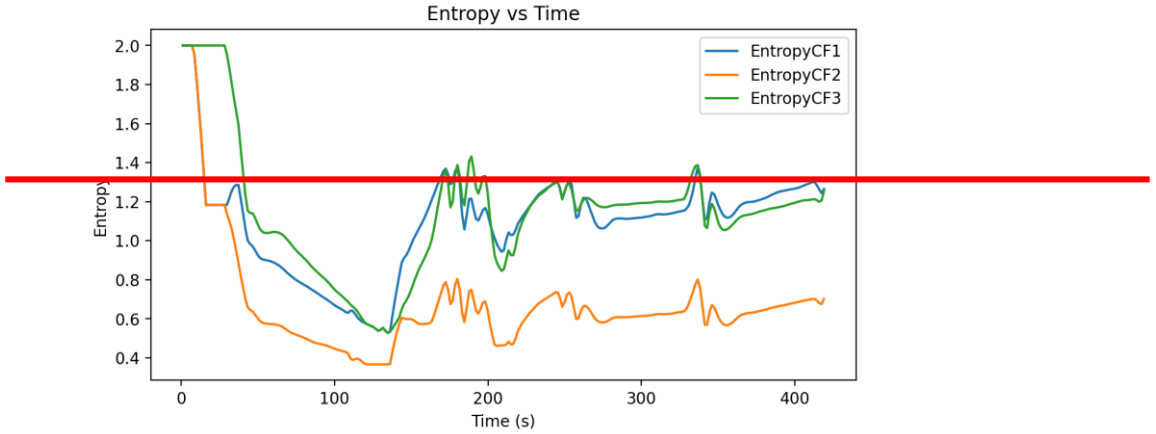
Dmin: 35cm
Dmax: **1.00m**
Threshold: 1.20



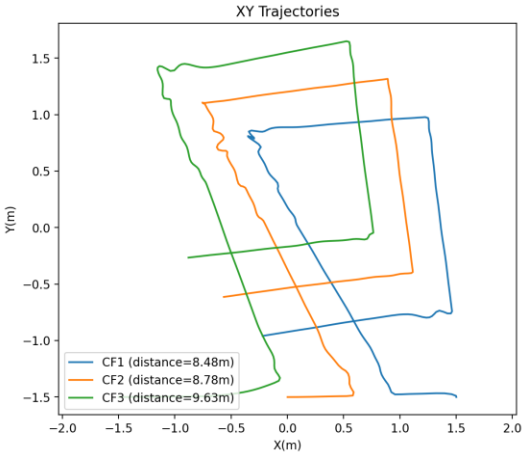


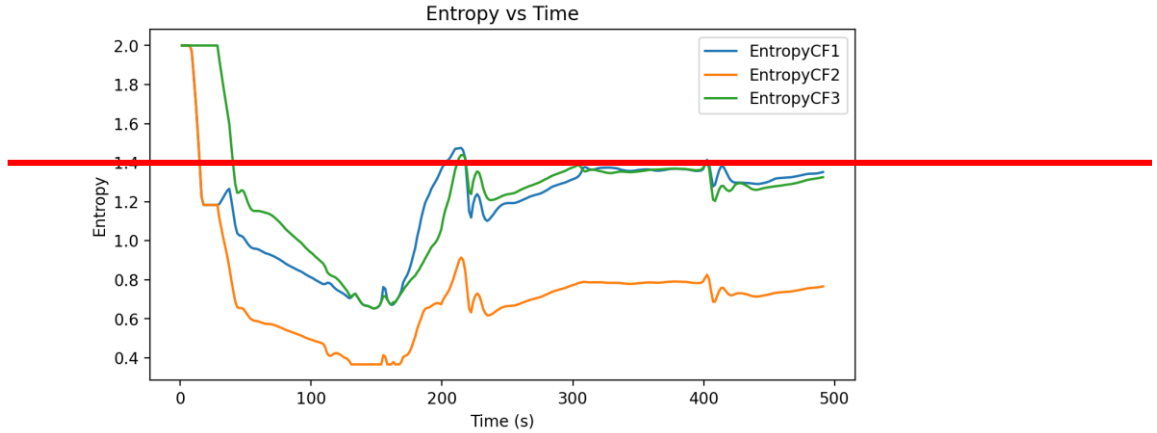
Dmin: 35cm
Dmax: **1.00m**
Threshold: 1.30



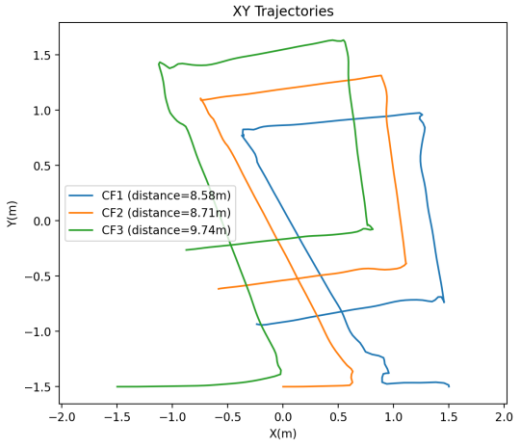


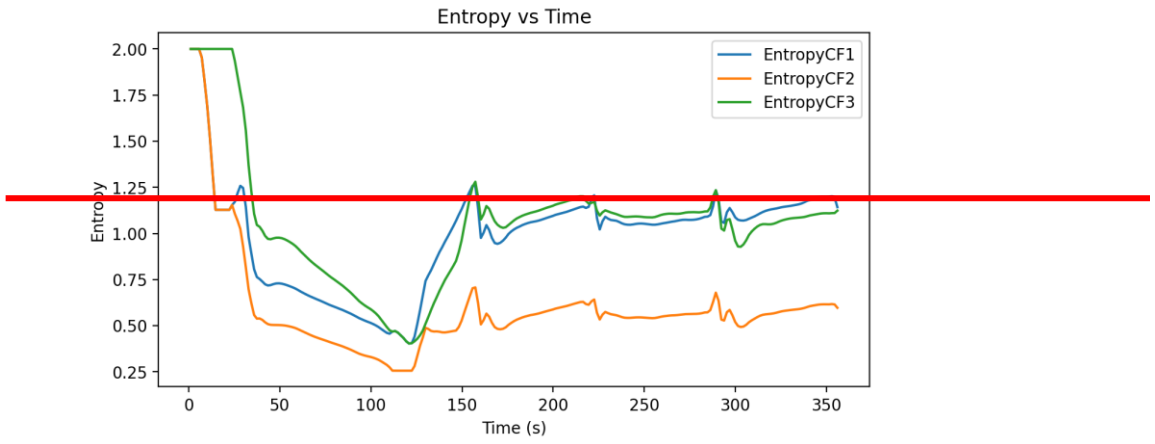
Dmin: 35cm
Dmax: **1.00m**
Threshold: 1.40



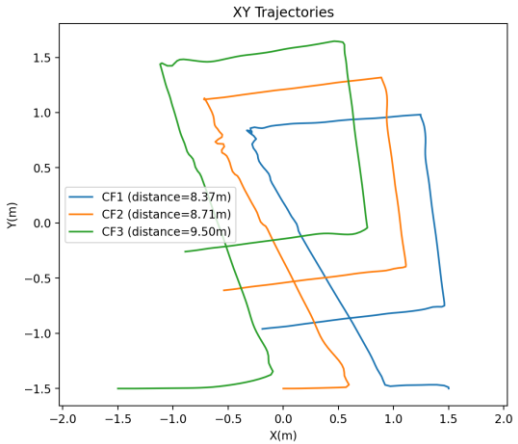


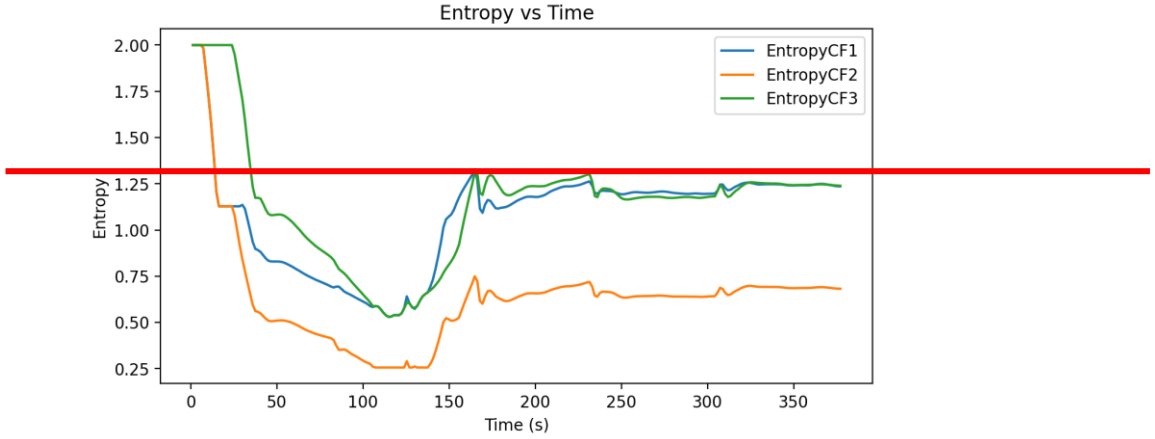
Dmin: 35cm
Dmax: **1.10m**
Threshold: 1.20



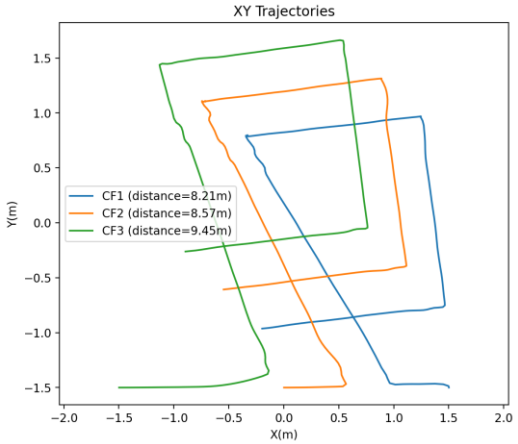


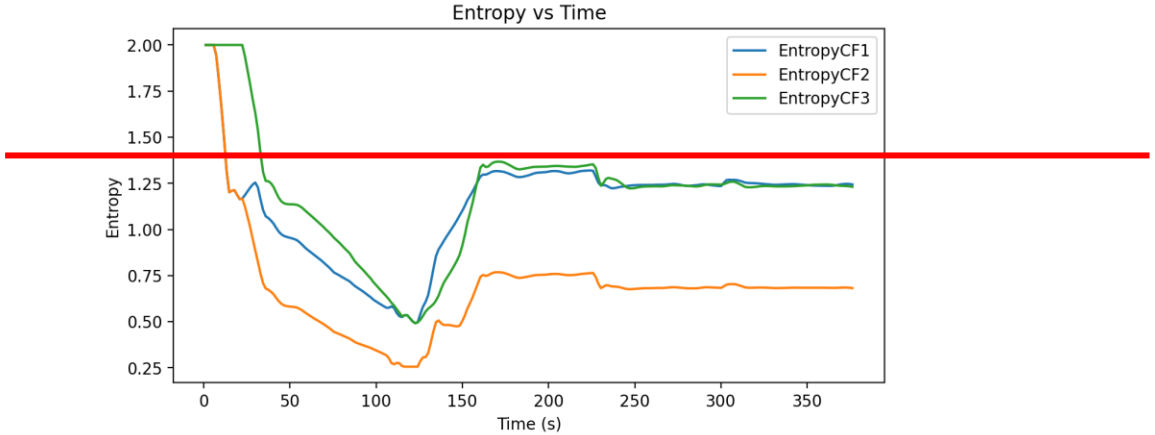
Dmin: 35cm
Dmax: **1.10m**
Threshold: 1.30



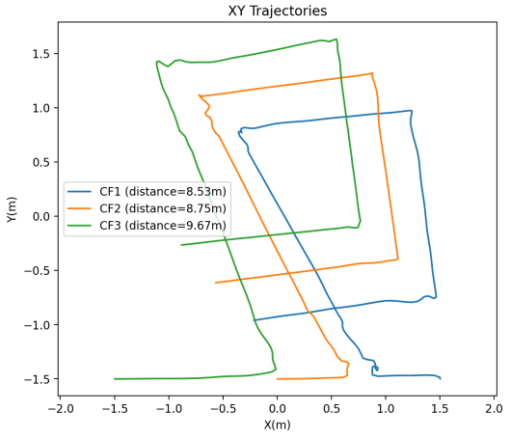


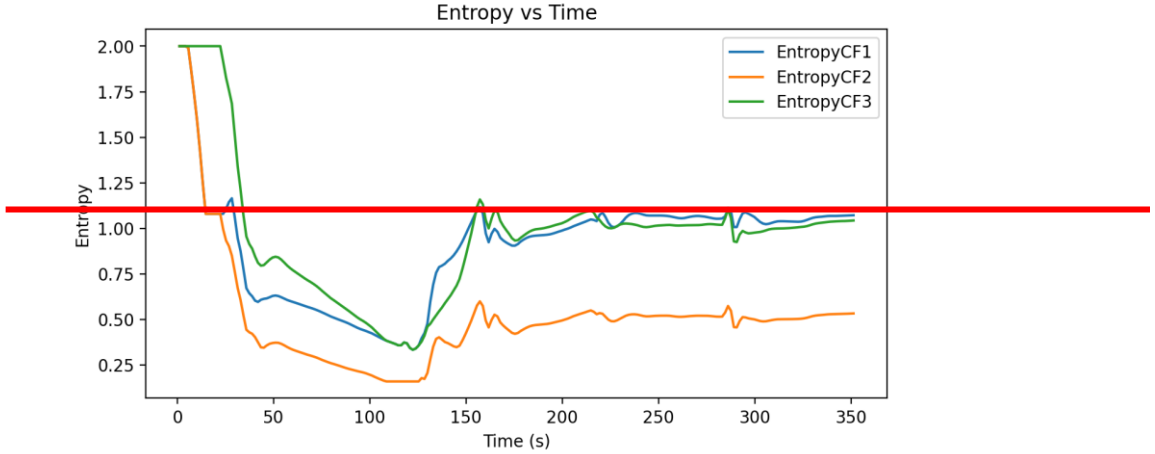
Dmin: 35cm
Dmax: **1.10m**
Threshold: 1.40



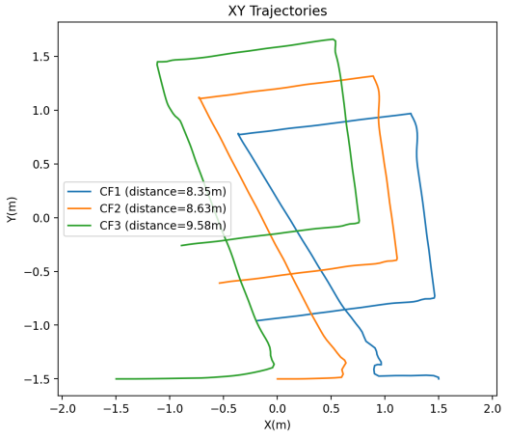


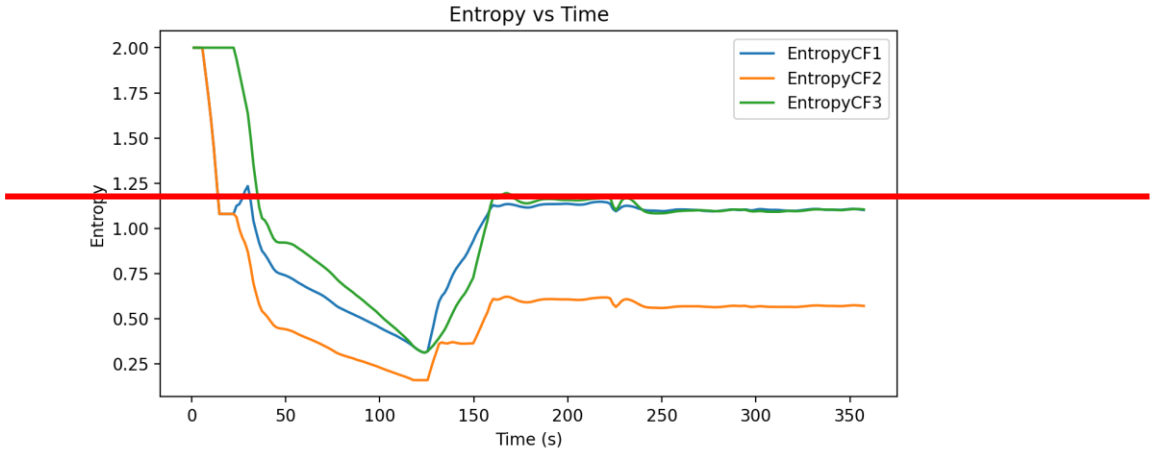
Dmin: 35cm
Dmax: **1.20m**
Threshold: 1.10



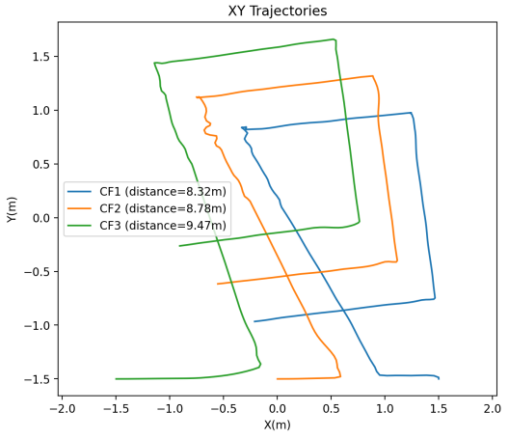


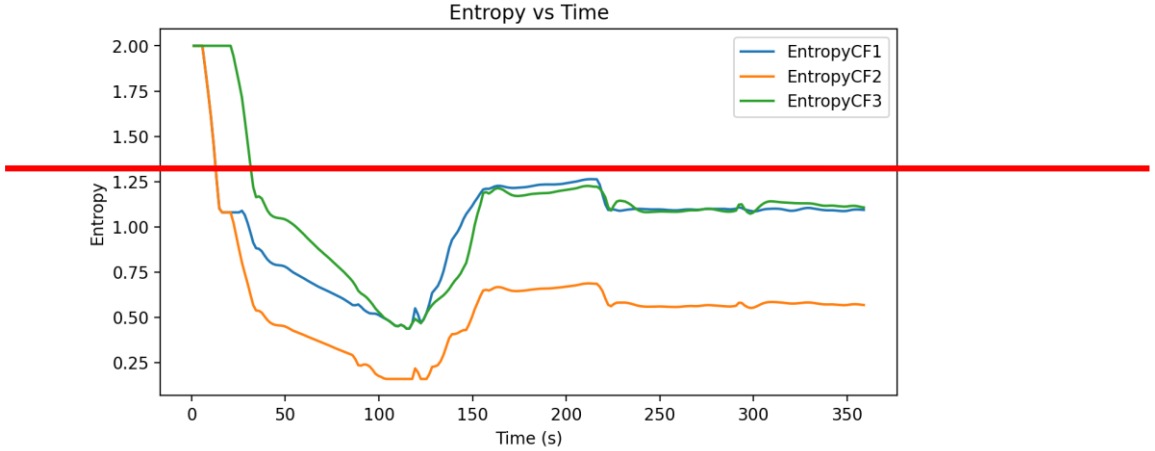
Dmin: 35cm
Dmax: **1.20m**
Threshold: 1.20





Dmin: 35cm
Dmax: **1.20m**
Threshold: 1.30

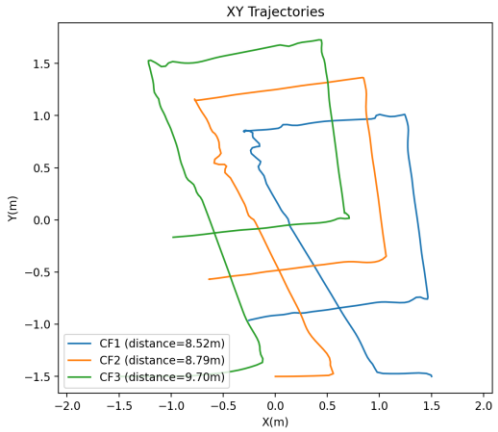


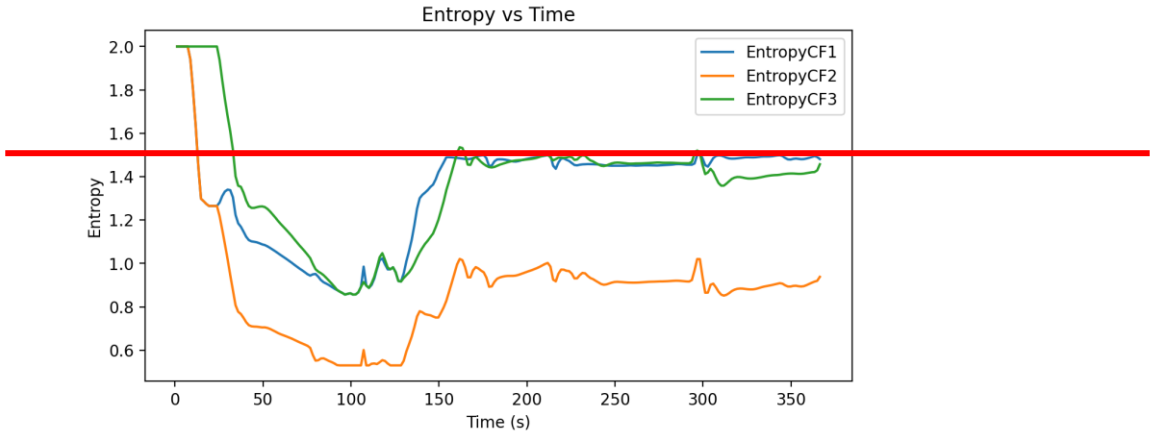


Dmin 0.40

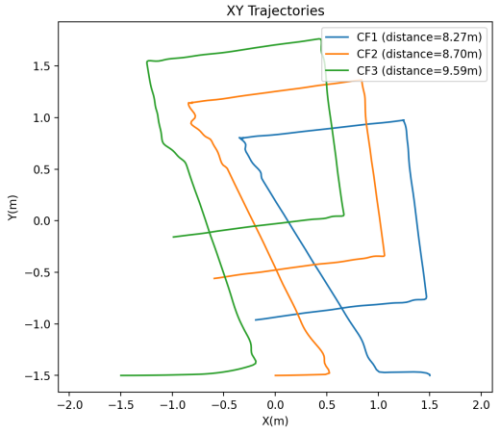
Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.40	1	1.4				x
		1.5	8.52	8.79	9.7	9.01
		1.6	8.27	8.7	9.59	8.85
		1.7	8.19	8.52	9.4	8.7
	1.1	1.2				X
		1.3	9	8.84	10.26	9.36
		1.4	8.39	8.76	9.93	9.03
		1.5	8.21	8.83	9.55	8.86
	1.2	1.2				X
		1.3	8.5	8.81	9.61	8.97
		1.4	8.23	8.81	9.51	8.85
		1.5	8.13	8.62	9.43	8.73

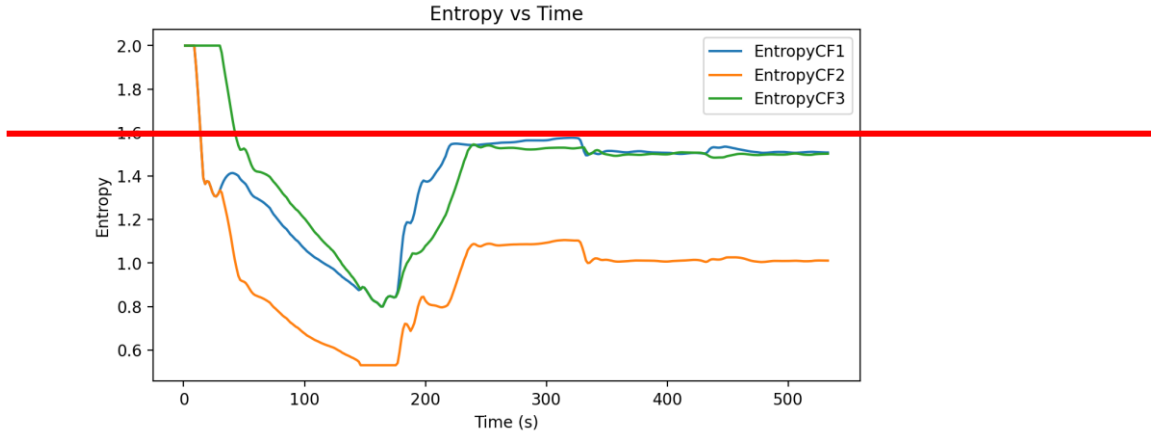
Dmin: 40cm
Dmax: **1.00m**
Threshold: 1.50



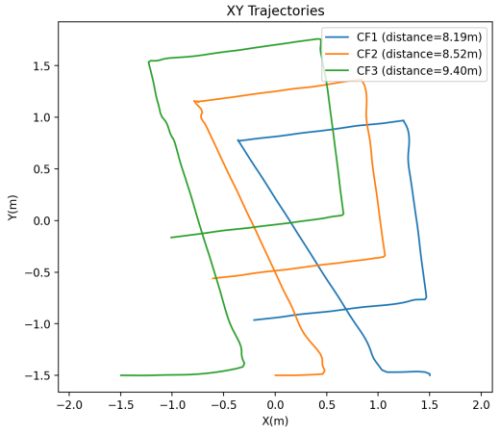


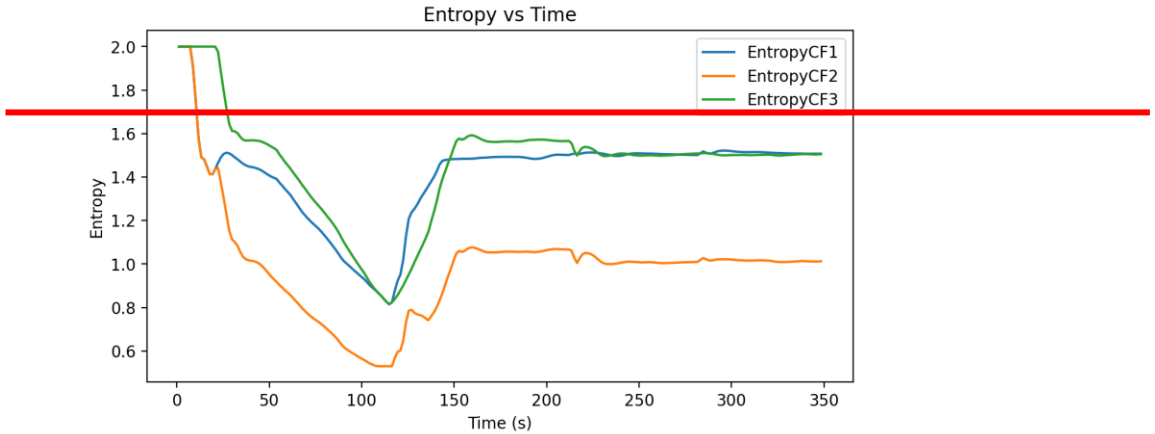
Dmin: 40cm
Dmax: **1.00m**
Threshold: 1.60



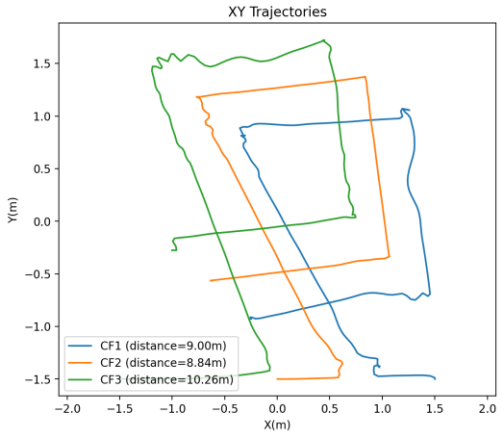


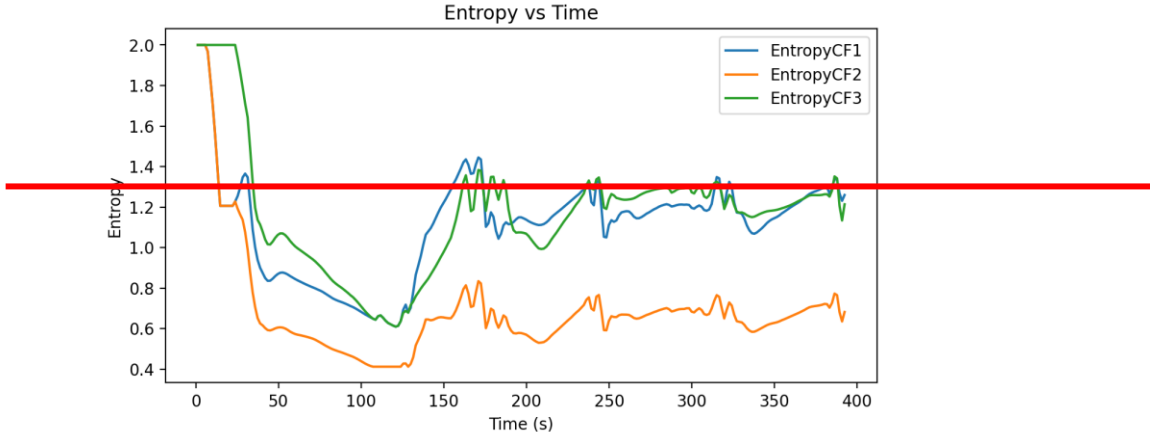
Dmin: 40cm
Dmax: **1.00m**
Threshold: 1.70



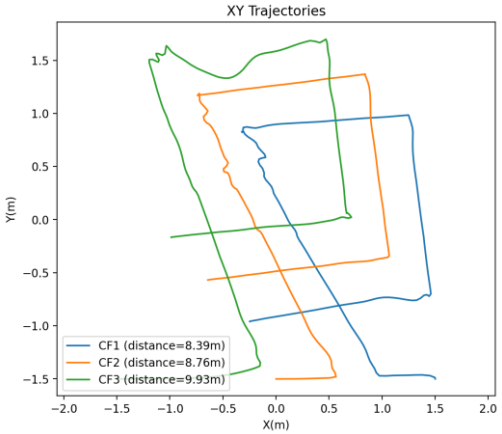


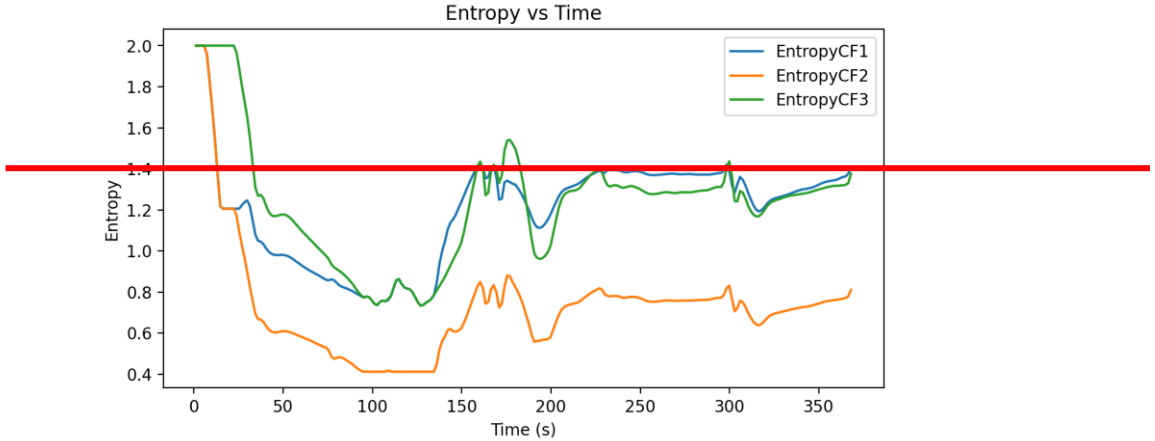
Dmin: 40cm
Dmax: **1.10m**
Threshold: 1.30



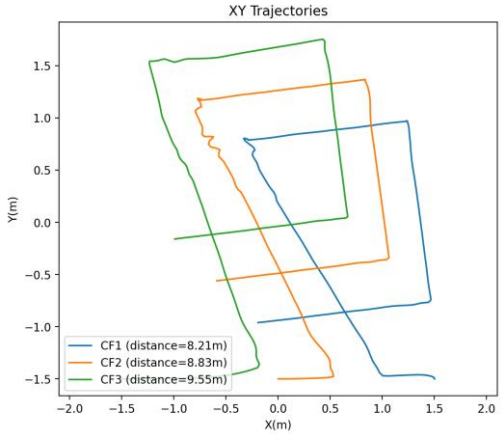


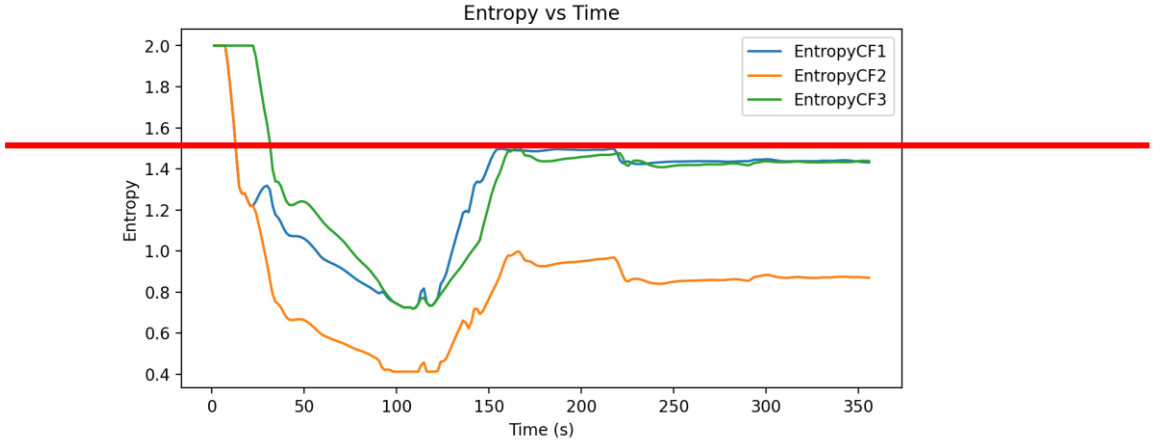
Dmin: 40cm
Dmax: **1.10m**
Threshold: 1.40



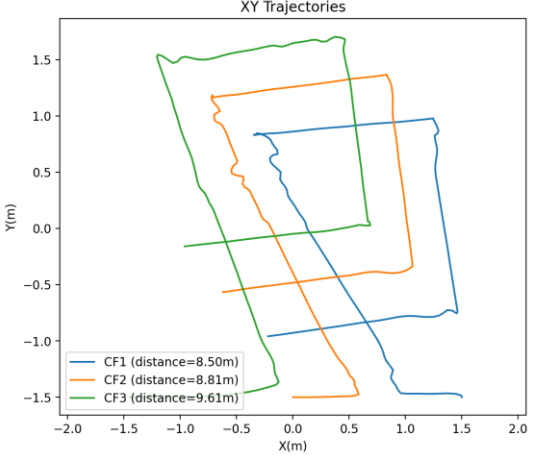


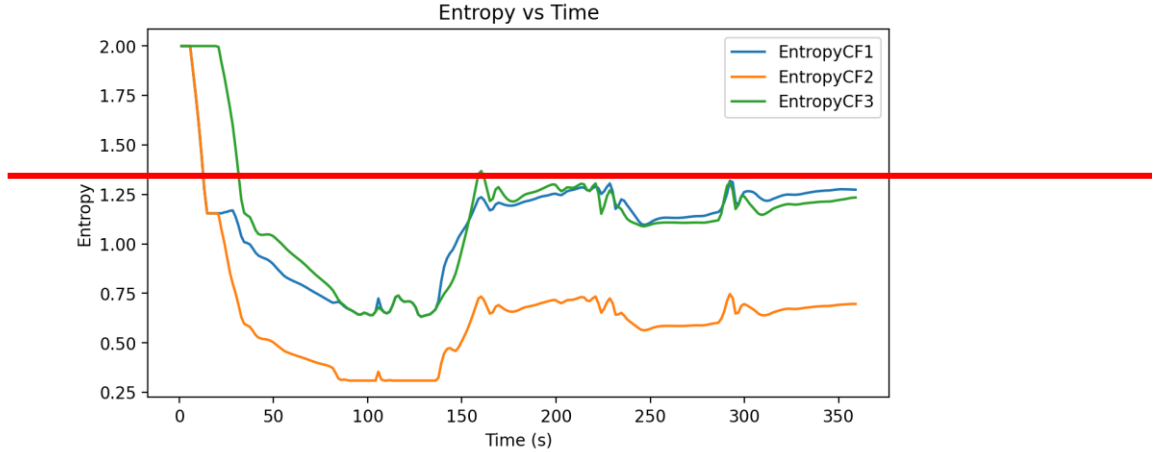
Dmin: 40cm
Dmax: **1.10m**
Threshold: 1.50



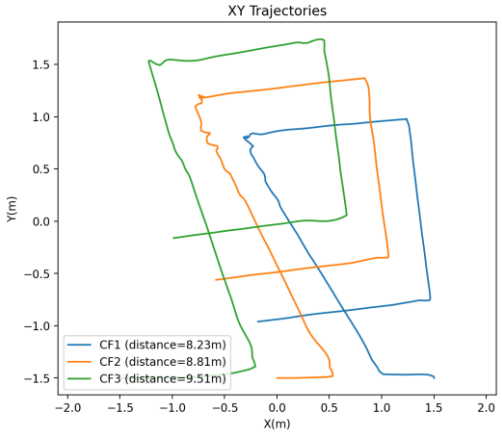


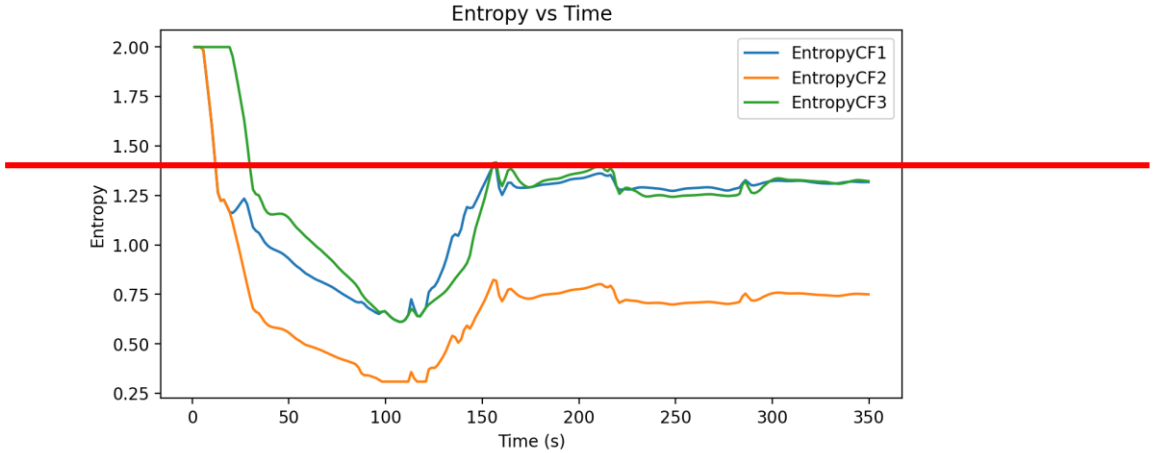
Dmin: 40cm
Dmax: **1.20m**
Threshold: 1.30



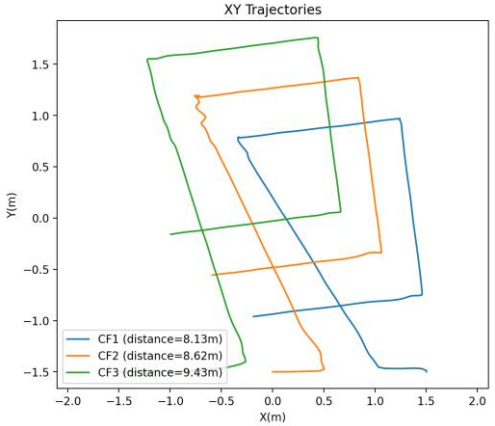


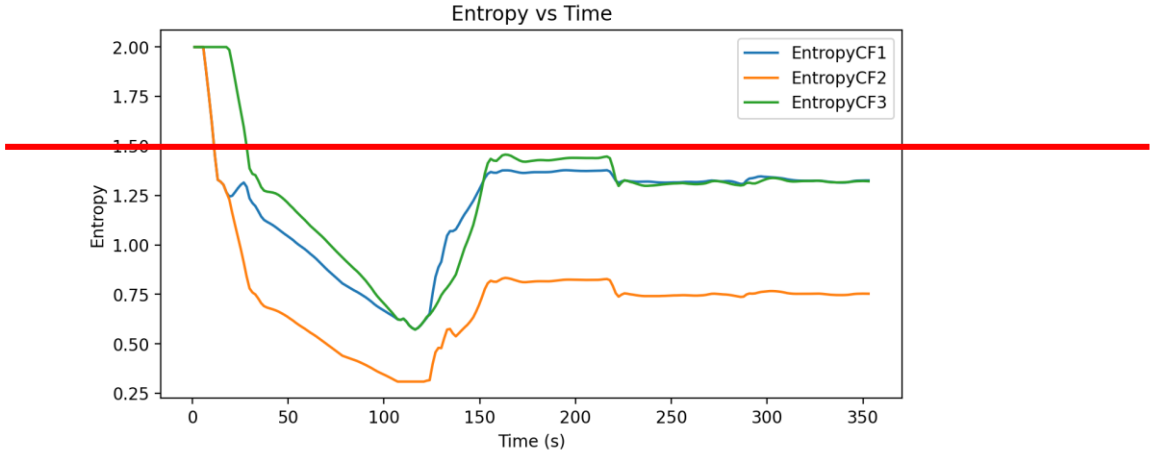
Dmin: 40cm
Dmax: **1.20m**
Threshold: 1.40





Dmin: 40cm
Dmax: **1.20m**
Threshold: 1.50





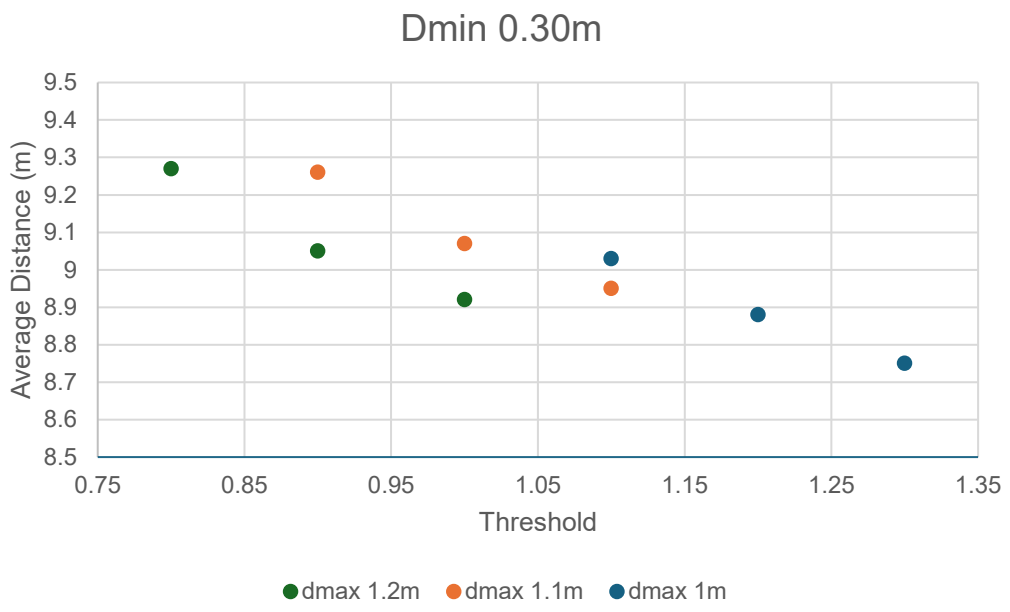
Results

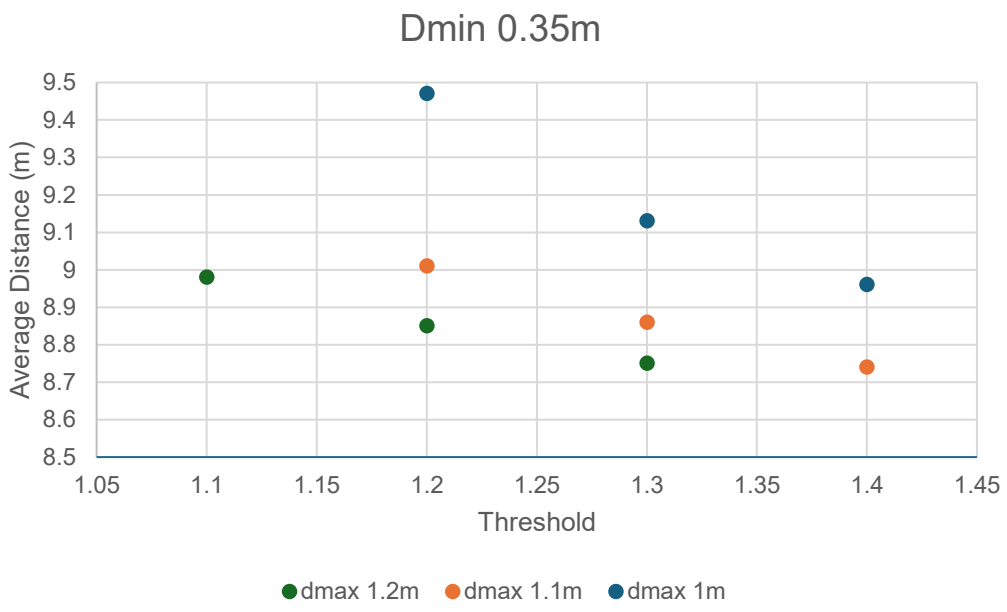
Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.3	1	1				X
		1.1	8.65	8.74	9.71	9.03
		1.2	8.4	8.61	9.62	8.88
		1.3	8.27	8.54	9.45	8.75
	1.1	0.8				x
		0.9	9.07	8.84	9.88	9.26
		1	8.75	8.75	9.71	9.07
		1.1	8.52	8.71	9.64	8.95
	1.2	0.7				X
		0.8	8.88	8.93	10	9.27
		0.9	8.71	8.76	9.69	9.05
		1	8.58	8.62	9.55	8.92

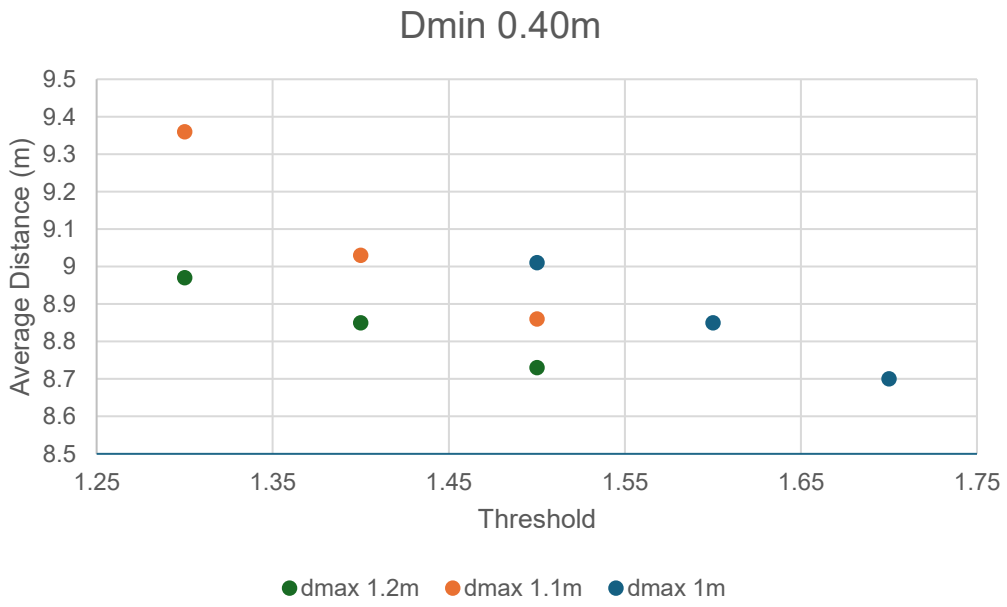
Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.35	1	1.1				x
		1.2	9.17	8.84	10.38	9.47
		1.3	8.57	8.7	10.14	9.13
		1.4	8.48	8.78	9.63	8.96
	1.1	1.1				X
		1.2	8.58	8.71	9.74	9.01
		1.3	8.37	8.71	9.5	8.86
		1.4	8.21	8.57	9.45	8.74
	1.2	1				X
		1.1	8.53	8.75	9.67	8.98
		1.2	8.35	8.63	9.58	8.85
		1.3	8.32	8.78	9.47	8.85

Dimin	Dmax	Threshold	Travel Distance CF1	CF2	CF3	Average
0.4	1	1.4				x
		1.5	8.52	8.79	9.7	9.01
		1.6	8.27	8.7	9.59	8.85
		1.7	8.19	8.52	9.4	8.7
	1.1	1.2				X
		1.3	9	8.84	10.26	9.36
		1.4	8.39	8.76	9.93	9.03
		1.5	8.21	8.83	9.55	8.86
	1.2	1.2				X
		1.3	8.5	8.81	9.61	8.97
		1.4	8.23	8.81	9.51	8.85
		1.5	8.13	8.62	9.43	8.73

Results


Dmin 0.30m
Average Distance (m)
9.5
9.4
9.3
9.2
9.1
9
8.9
8.8
8.7
8.6
8.5
0.75
0.85
0.95
1.05
1.15
1.25
1.35
Threshold
dmax 1.2m
dmax 1.1m
dmax 1m


Dmin 0.35m
Average Distance (m)
9.5
9.4
9.3
9.2
9.1
9
8.9
8.8
8.7
8.6
8.5
1.05
1.1
1.15
1.2
1.25
1.3
1.35
1.4
1.45
Threshold
dmax 1.2m
dmax 1.1m
dmax 1m


Dmin 0.40m
Average Distance (m)
9.5
9.4
9.3
9.2
9.1
9
8.9
8.8
8.7
8.6
8.5
1.25
1.35
1.45
1.55
1.65
1.75
Threshold
dmax 1.2m
dmax 1.1m
dmax 1m